A Floor-Saving Lower Bound for Unique Positive Differences

Proof sketch

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All logarithms are natural, and $\mathbb{N} = \{1, 2, 3, \dots\}$. Let

$$\lambda = \frac{1}{2\log 2}, \qquad \gamma = \lim_{N\to\infty}\left(\sum_{m\le N}\frac{1}{m} - \log N\right), \qquad B_1 = \frac{1}{2} + \gamma - \log(\log 2).$$

Theorem 1. *Let $A \subseteq \mathbb{N}$ be such that, for every $n \ge 1$, there is exactly one ordered pair $(a, b) \in A^2$ with $a > b$ and $n = a - b$. Write this pair as (a_n, b_n). For every fixed $B > B_1$, arbitrarily large n satisfy*

$$a_n > \lambda\frac{n^2}{\log n - \log\log n + B}.$$

Consequently,

$$\limsup_{n\to\infty}\frac{a_n(\log n - \log\log n + B_1)}{n^2} \ge \lambda.$$

Sketch. For a representation $n = t - b$, call t the *top*. The spine is

$$\text{spacing} \longrightarrow \text{top count} \longrightarrow \text{integer floor} \longrightarrow \text{constant comparison}.$$

The continuous majorant has the correct main term X. The proof wins at the next scale: the integer floor subtracts a deterministic fractional-part mass of order $X/\log X$.

1. Contradiction setup and inverse weight. Fix $B > B_1$ and suppose that eventually

$$a_n \le \Phi(n), \qquad \Phi(x) = \lambda\frac{x^2}{H(\log x)}, \qquad H(w) = w - \log w + B. \tag{1}$$

Choose $N_0 \ge 2$ so that (1) holds for $n \ge N_0$ and Φ is increasing on $[N_0, \infty)$. Put $T_* := \Phi(N_0)$, let ψ be the inverse of this branch, and set $f(t) = 1/\psi(t)$ for $t \ge T_*$, extending f constantly on $[0, T_*]$. Then f is positive and nonincreasing. Inverting $t = \lambda r^2/H(\log r)$ gives

$$f(t) = \sqrt{\frac{2\lambda}{t\log t}}(1 + o(1)), \qquad I(T) := \int_0^T f(t)\,dt = 2\sqrt{\frac{2\lambda T}{\log T}} + o\left(\sqrt{\frac{T}{\log T}}\right). \tag{2}$$

All o-terms below are for $X \to \infty$, with A and B fixed.

2. Spacing. Choose an integer

$$Z_0 > \max\left(T_*, \max_{1\le d<N_0} a_d\right).$$

If $y < z$ are elements of A and $z > Z_0$, then (z, y) represents $d = z - y$. Uniqueness forces $z = a_d$. The choice of Z_0 excludes $d < N_0$, so (1) gives $z \le \Phi(d)$ and hence $d \ge \psi(z)$. Since f is nonincreasing,

$$\int_y^z f(s)\,ds \ge (z - y)f(z) = d/\psi(z) \ge 1. \tag{3}$$

Thus every large gap in A costs one unit of f-mass. Summing the gaps gives

$$|A \cap [1, T]| \le I(T) + O(1). \tag{4}$$

We also need the weighted form. If $z_1 < \cdots < z_L$ are elements of A above Z_0 and $W \ge 0$ is nonincreasing, then

$$\sum_{j=2}^{L} W(z_j) \le \int_{z_1}^{z_L} f(t)W(t)\,dt. \tag{5}$$

*Large language model by OpenAI; assisted with derivation and drafting through prompted interaction.

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Indeed, apply (3) to each gap $[z_{j-1}, z_j]$ and use $W(t) \geq W(z_j)$ on that gap.

3. Top count and floor. Let X be a large integer, put $Y = \log X$ and $M = \Phi(X)$. For all large X, every difference $1 \leq n \leq X$ has top at most M: use (1) for $n \geq N_0$; the finitely many $n < N_0$ have fixed tops, eventually below M. Hence

$$X = \sum_{t \in A:\, t \leq M} r_t, \qquad r_t := \#\{b \in A : t - X \leq b < t\}. \tag{6}$$

This identity is exact: a counted pair gives a difference in $[1, X]$, and uniqueness prevents double counting. The terms with $t \leq X$ are exactly $\binom{|A \cap [1,X]|}{2}$, so (4) gives

$$\sum_{t \in A:\, t \leq X} r_t \leq \frac{1}{2} I(X)^2 + o(X/Y). \tag{7}$$

For $t > X$ define

$$G_X(t) = \int_{t-X}^{t} f(s)\, ds.$$

Since f is nonincreasing, G_X is nonincreasing. If $t \in A$ and $t > X + Z_0$, list the bottoms counted by r_t as $b_1 < \cdots < b_{r_t}$ and put $b_{r_t+1} = t$. Each upper endpoint in the gaps $[b_i, b_{i+1}]$ is above Z_0, and these gaps lie in $[t - X, t]$. Hence (3) gives

$$r_t \leq \sum_{i=1}^{r_t} \int_{b_i}^{b_{i+1}} f(s)\, ds \leq G_X(t).$$

Since r_t is an integer,

$$r_t \leq \lfloor G_X(t) \rfloor. \tag{8}$$

This is the only non-continuous step. Apply (5) to the large tops with $W(t) = \lfloor G_X(t) \rfloor$. The first large top has at most $I(X) + O(1)$ bottoms, and the fixed strip $X < t \leq X + Z_0$ has $O(1)$ possible tops, each with $O(I(X))$ bottoms. By (2), these losses are $o(X/Y)$. Therefore

$$X \leq \frac{1}{2} I(X)^2 + \int_X^M f(t) \lfloor G_X(t) \rfloor\, dt + o(X/Y). \tag{9}$$

Equivalently,

$$X \leq J_B(X) - E(X) + o(X/Y), \tag{10}$$

where

$$J_B(X) = \frac{1}{2} I(X)^2 + \int_X^M f(t) G_X(t)\, dt, \qquad E(X) = \int_X^M f(t) \{G_X(t)\}\, dt.$$

4. Continuous count. The continuous majorant is

$$J_B(X) = X + \lambda(3 - 2B - 2\log(\log 2)) \frac{X}{Y} + o(X/Y). \tag{11}$$

Here is the constant ledger. First,

$$\frac{1}{2} I(X)^2 = 4\lambda X/Y + o(X/Y). \tag{12}$$

Next split

$$\int_X^M f(t) G_X(t)\, dt = X \int_X^M f(t)^2\, dt + R(X),$$

where

$$R(X) = \int_X^M f(t) \int_{t-X}^{t} (f(s) - f(t))\, ds\, dt.$$

For the f^2 term use $t = \Phi(e^w)$ and write $w_0 = \log \psi(X)$. Then

$$w_0 = Y/2 + (\log Y)/2 - (\log(2\lambda))/2 + o(1), \qquad M = \Phi(e^Y).$$

Also $f(\Phi(e^w)) = e^{-w}$ and $dt = \Phi'(e^w) e^w\, dw$, so

$$f(t)^2\, dt = \lambda \left(\frac{2}{H(w)} - \frac{H'(w)}{H(w)^2} \right) dw.$$

Thus

$$\int_X^M f(t)^2\,dt = \lambda\left[2\int_{w_0}^Y \frac{dw}{H(w)} + \frac{1}{H(Y)} - \frac{1}{H(w_0)}\right]. \tag{13}$$

Writing $w_0 = Y/2 + \rho_X$, the three $1/Y$ contributions inside the bracket are

$$\begin{aligned} 2\int_{w_0}^Y \frac{dw}{w} &= 2\log 2 - \frac{4\rho_X}{Y} + o(1/Y), \\ 2\int_{w_0}^Y \frac{\log w - B}{w^2}\,dw &= \frac{2\log Y - 4\log 2 + 2 - 2B}{Y} + o(1/Y), \\ \frac{1}{H(Y)} - \frac{1}{H(w_0)} &= -\frac{1}{Y} + o(1/Y). \end{aligned} \tag{14}$$

Since $\rho_X = (\log Y - \log(2\lambda))/2 + o(1)$, the logarithms cancel and, using $2\lambda\log 2 = 1$,

$$X\int_X^M f(t)^2\,dt = X + \lambda(1 - 2B + 2\log(\lambda/2))\frac{X}{Y} + o(X/Y). \tag{15}$$

For $R(X)$, put $t = Xu$ and $s = Xv$. The local form of (2) on bounded u-ranges, followed by truncation, gives

$$R(X) = \frac{2\lambda X}{Y}\int_1^\infty \left(\int_{u-1}^u \frac{dv}{\sqrt{uv}} - \frac{1}{u}\right)du + o(X/Y) = 2\lambda(2\log 2 - 1)X/Y + o(X/Y). \tag{16}$$

Adding (12), (15), and (16), and using $4\log 2 + 2\log(\lambda/2) = -2\log(\log 2)$, gives (11).

5. Floor saving. The floor subtracts

$$E(X) = 2\lambda(1-\gamma)X/Y + o(X/Y). \tag{17}$$

Set

$$q(t) = Xf(t) = X/\psi(t).$$

The range $\Phi(X/Q) \le t \le M$ corresponds to $1 \le q \le Q$. With $r = \psi(t) = X/q$, for every measurable $|F_X| \le 1$,

$$\begin{aligned} \int_{\Phi(X/Q)}^M f(t)F_X(q(t))\,dt &= X\int_1^Q \frac{\Phi'(X/q)}{X/q}F_X(q)\frac{dq}{q^2} \\ &= \frac{2\lambda X}{Y}\int_1^Q F_X(q)\frac{dq}{q^2} + o_Q(X/Y). \end{aligned} \tag{18}$$

Also, uniformly for $1 \le q \le Q$,

$$G_X(\Phi(X/q)) = q + o_Q(1), \tag{19}$$

because f varies by $o_Q(1)$ relatively on the interval of length X ending at $\Phi(X/q)$, while $Xf(\Phi(X/q)) = q$ exactly.

Apply (18) with $F_X(q) = \{G_X(\Phi(X/q))\}$. By (19), the integrand converges to $\{q\}$ away from the finitely many integer discontinuities in $[1, Q]$; small neighborhoods of those discontinuities have arbitrarily small $q^{-2}\,dq$ mass. Thus the saving is deterministic convergence under the natural q-measure, not an equidistribution assertion. The range $q > Q$ has mass only $O(X/(QY))$, since

$$\int_X^{\Phi(X/Q)} f(t)\,dt = \int_{\psi(X)}^{X/Q} \frac{\Phi'(r)}{r}\,dr \ll \frac{X}{QY}.$$

Letting $X \to \infty$ and then $Q \to \infty$,

$$E(X) = \frac{2\lambda X}{Y}\int_1^\infty \frac{\{q\}}{q^2}\,dq + o(X/Y).$$

Finally,

$$\int_1^\infty \frac{\{q\}}{q^2}\,dq = \sum_{m\ge 1}\int_m^{m+1}\frac{q-m}{q^2}\,dq = \sum_{m\ge 1}\left(\log\frac{m+1}{m} - \frac{1}{m+1}\right) = 1-\gamma,$$

which proves (17). On the q-scale the measure is $(2\lambda X/Y)q^{-2}\,dq$, and the floor removes the average fractional part $1-\gamma$.

6. Contradiction. Combining (10), (11), and (17),

$$\begin{aligned} X &\le X + \lambda(3 - 2B - 2\log(\log 2))X/Y - 2\lambda(1-\gamma)X/Y + o(X/Y) \\ &= X - 2\lambda(B - B_1)X/Y + o(X/Y). \end{aligned}$$

Since $B > B_1$, this is impossible for large X. Thus (1) cannot hold eventually.

For the endpoint limsup, apply the conclusion with any fixed $B^* > B_1$ along an unbounded sequence n_j. Then

$$\frac{a_{n_j}(\log n_j - \log\log n_j + B_1)}{n_j^2} > \lambda \frac{\log n_j - \log\log n_j + B_1}{\log n_j - \log\log n_j + B^*} \to \lambda,$$

so the limsup is at least λ. $\square$